

# Ramana Gangarao

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## Experience

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### AI Engineer

*Innovapath INC*

*Dublin, CA*

*Jan 2025 – present*

- Built an enterprise-grade agentic AI internal workforce operations platform automating different workflows via conversational AI.
- Designed a multi-agent system with centralized orchestration, intent routing, tool invocation, and coordinated task execution across domain-specific agents.
- Architected a secure, multi-tenant AI platform integrating internal enterprise systems and tools with strong data isolation across users and organizations.
- Used Google Agent Development Kit (Google ADK) as the primary agentic AI framework for deterministic orchestration and agent lifecycle management.
- Built persistent state, session, and memory management layers using durable storage and caching to support long-running, multi-turn conversations.
- Developed structured prompting, prompt versioning, and regression testing workflows to ensure consistent, policy-aligned, and reliable agent behavior.
- Implemented RAG pipelines using Milvus vector database for low-latency semantic retrieval over large-scale unstructured enterprise documents.
- Designed evaluation pipelines using Langfuse and custom metrics to assess agent accuracy, grounding quality, latency, and failure modes.
- Productionized the system using Dockerized services deployed on AWS EKS with Bedrock inference, S3 storage, and scalable agent workloads.
- Implemented end-to-end observability with CloudWatch, enabling agent traceability, tool execution logs, latency monitoring, and audit-ready telemetry.

### Developer Analyst I / MLOPS

*Wipro Limited*

*Chennai, India*

*Sept 2021 – Aug 2024*

- Architected an enterprise-wide feature platform by defining versioned feature views, schemas, TTLs, and ownership metadata in AWS Glue, enabling governed and reusable ML assets across product teams.
- Built durable batch + streaming ingestion by routing event/file sources into S3 with schema registration and lineage capture, ensuring reproducible, audit-ready flows for regulated environments.
- Developed deterministic, idempotent ETL pipelines with data-quality enforcement (schema, null, distribution checks) to generate high-fidelity feature vectors powering multiple high-traffic ML systems.
- Engineered offline feature tables using scheduled Spark jobs to compute windowed aggregations with point-in-time correctness and partitioning, improving training dataset generation performance.
- Delivered low-latency online serving by streaming incremental updates into a KV store with atomic upserts and TTL enforcement, maintaining cross-environment feature parity for real-time models.
- Published unified SDKs and REST/gRPC APIs with built-in lineage, freshness telemetry, and drift/error metrics, streamlining feature discovery and adoption for data scientists and ML engineers.
- Built Airflow-orchestrated training pipelines on Databricks that version datasets, run distributed training and hyperparameter sweeps, and register lineage-complete artifacts in the model registry.
- Implemented a multi-stage model promotion pipeline (staging → canary → prod) with CI/CD, drift checks, benchmark validation, and policy/compliance gates, reducing deployment risk and audit overhead.
- Built a dynamic API routing layer via API Gateway with authentication, tenant isolation, A/B and shadow traffic routing, and instant rollback, enabling safe experimentation and rapid iteration.

- Designed scalable batch inference workflows on SageMaker Batch to produce population-level predictions with idempotent outputs and full dataset versioning.
- Established end-to-end ML observability using Prometheus and CloudWatch, with custom monitors to track latency, error rates, feature drift, prediction stability, and data skew for both real-time and batch models.
- Implemented automated drift and anomaly detection using PSI/KL metrics, input-output validators, and PagerDuty/Slack alerting to trigger retraining or rollback workflows and maintain continuous model health.

## Software Engineer

INNOVAPATH IT Solutions PVT Ltd

Hyderabad, India

Sept 2016 – Sept 2021

- Built and deployed Spring Boot microservices using JPA/Hibernate, applying design patterns and Java 8 features for scalable backend development.
- Developed automation frameworks using Selenium, TestNG, RESTAssured, and BDD, improving UI and API test coverage and maintainability.
- Containerized Java and Python services with Docker and deployed on AWS EC2 and S3, ensuring high availability and production stability.
- Implemented CI/CD pipelines using Jenkins and Git, automating builds, tests, and deployments for faster and reliable software delivery.

## Skills

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**Programming Languages:** Python, JavaScript, Java

**Frameworks:** Scikit-learn, NumPy, Pandas, Pytorch, MLflow, Reactjs, Nodejs, Redux

**Libraries:** Google ADK, Langchain, LangGraph, Llamaindex, Langfuse , HuggingFace, Transformers

**LLMs:** Anthropic Claude, BERT, OpenAI

**Database:** SQL, PostgreSQL, MongoDB, MilvusDB, Neo4j

**AWS:** EKS, Bedrock, RDS, Lambda, CloudWatch, S3 bucket, EC2, API Gateway